# Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Supervised by Dr. Yihao Ding · Dr. Daochang Liu




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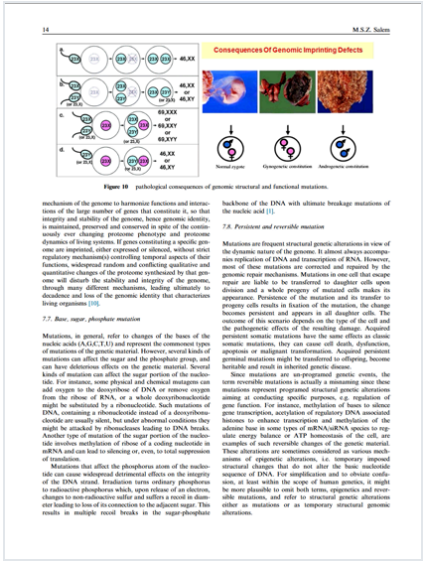
# *Table of Contents*


- **Introduction**
- Survey of MP-VRDU
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- Proof of Concept: A Simple Method

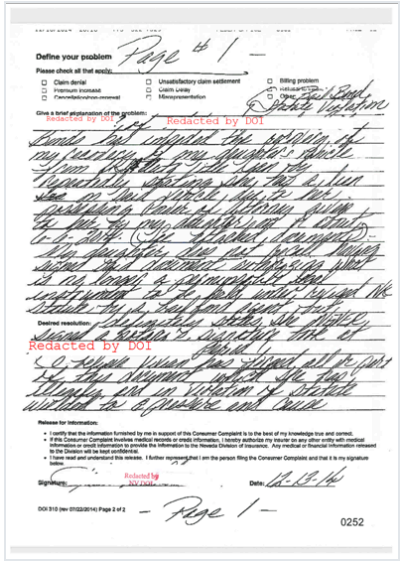

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# *Multi-Page Visually Rich Documents*

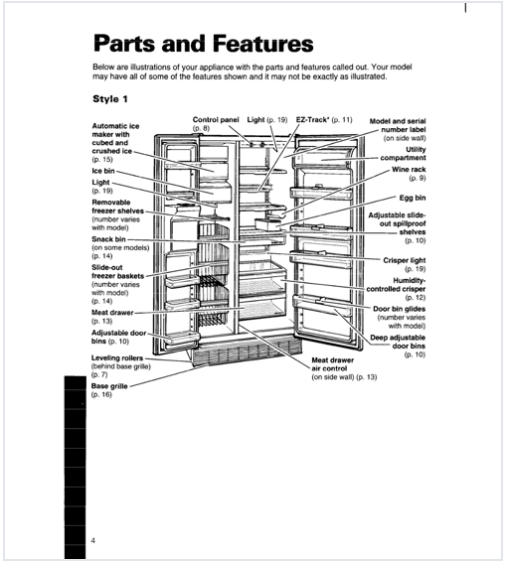
Documents carry meaning through **text**, **visual elements** like charts and figures, and **how they are laid out** on a page. In a **multi-page** document, meaning also connects across pages: a paragraph may refer to a figure several pages away.



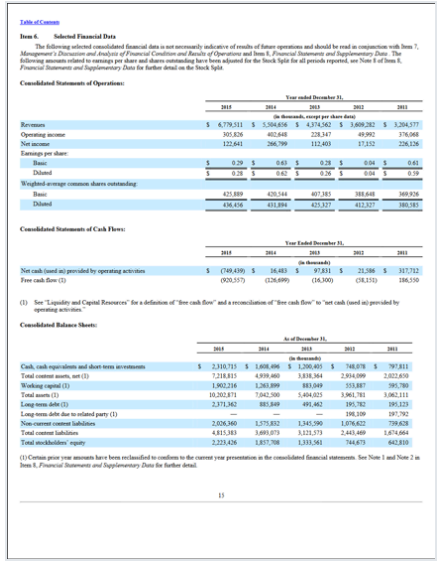

Research paper
born-digital




Signed form
scanned, handwritten




Technical manual
layout-dependent



Financial report
dense tables




Product brochure
photo-led

- **Text.** Typeset or handwritten, embedded in the file or only visible in a scan.
- **Visual information.** Charts, figures, photographs and handwriting.
- **Layout.** How information is arranged in the two-dimensional space of a page.

**Next** *This brings us to document understanding over long documents.*

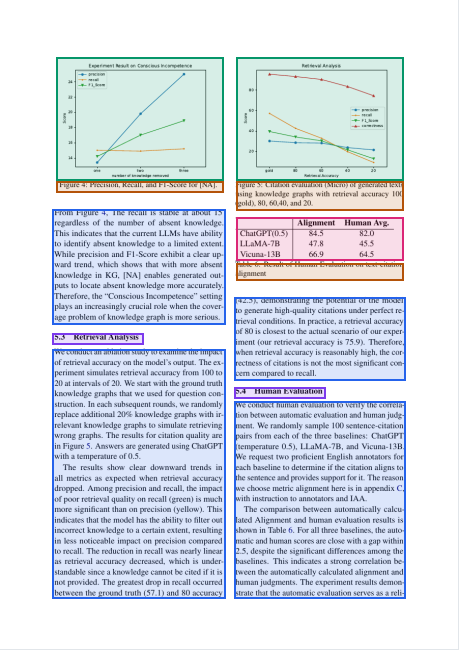
# *What is Long Document Understanding?*

Here: **answering a natural-language question** about a long, multi-page document. Difficult for two reasons.

**Volume**: hundreds of pages, costly to read, may not fit one context window. **Sparse evidence**: scattered across pages, to be located and connected.

LAYOUT ANALYSIS



DOCUMENT OUTLINE

**Research paper, 24 pages**

- 5.2 Conscious Incompetence
  - Paragraphs
  - Figure 4 — Caption
- 5.3 Retrieval Analysis
  - Paragraphs
  - Figure 5 — Caption
- 5.4 Human Evaluation
  - Paragraphs
  - Table 6 — Caption

EXAMPLE QUESTION

"Does the model rated best by humans in Table 6 also have the best citation recall in Table 3?"

Evidence needed:

- Table 6, this page
- Table 3, previous page

**Cross-page**: it compares two tables on different pages, so neither alone is enough.

Title · Section heading · Text · Figure / chart · Table · Caption / note · Header / footer

**Next** *With that in mind, the survey: the different approaches to document understanding.*

# *Table of Contents*


- Introduction
- **Survey of MP-VRDU**

  *Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding*

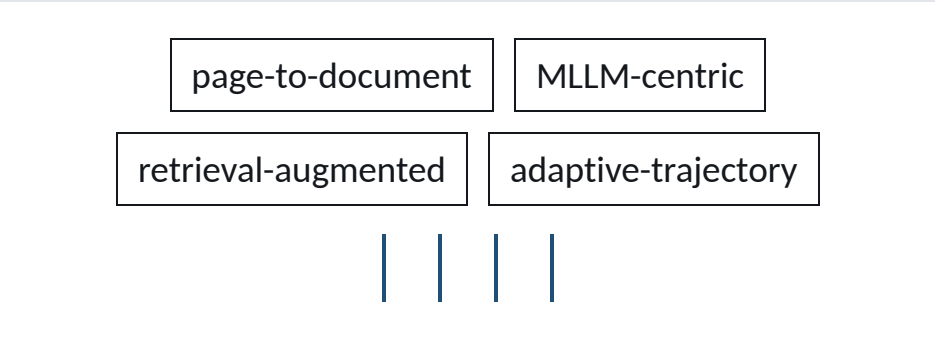
  Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

  **EMNLP 2026, main conference**
- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method


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# *Survey: Research Gap and Contributions*

## RESEARCH GAP




Multi-page systems developed along **parallel lines**, with different terminology and little comparison between them.

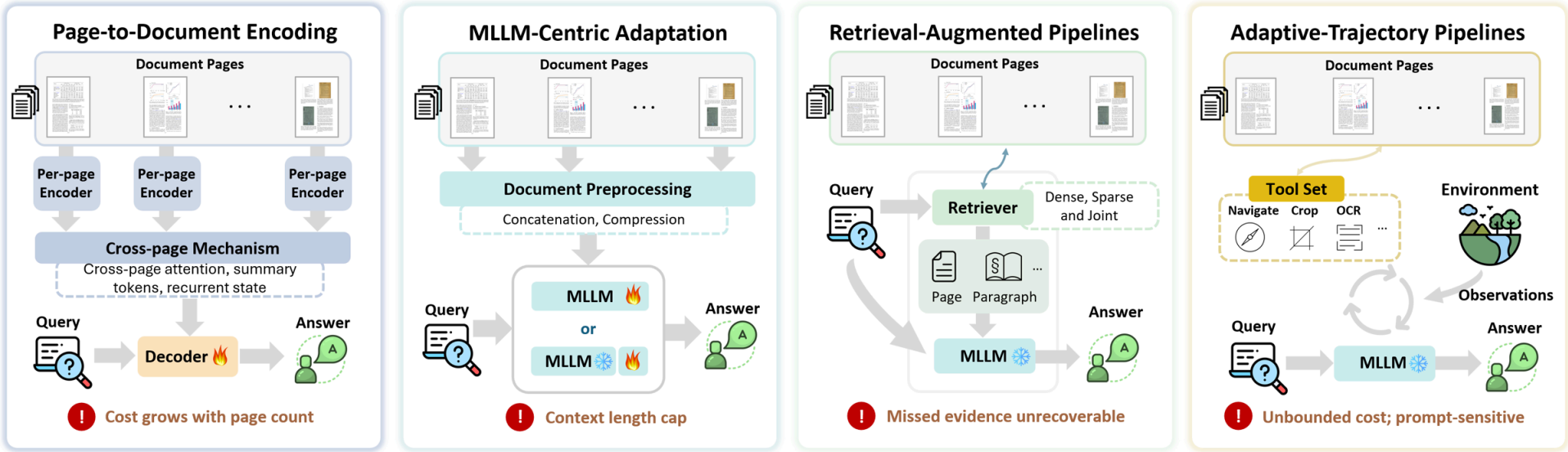
## CONTRIBUTIONS

1. An **evidence-management** perspective: how systems preserve, find and use evidence across a document.
2. A taxonomy of **four architectural families**.
3. **Multimodal representation** under document-length constraints: text, visual and structure.
4. **Inference-time and training strategies** for cross-page evidence: retrieval, reasoning, agentic methods.
5. Consolidates **datasets** and evaluation practice.
6. Five **open challenges**: evidence pipelines, faithfulness, efficiency, supervision integrity, evaluation.

**Next** *I will focus on the four families today.*

# *Multi-Page Architectural Overview*

Four architectural families, each motivated by the limits of the one before.




## Late 2023 to mid 2024

Each page is encoded into a compact representation, then connected by a cross-page mechanism. A fixed budget per page can lose fine detail.

## Early 2024 to late 2025

A pretrained multimodal model reads the document within its context window. Performance drops as documents grow, and some do not fit.

## Early 2024 to mid 2026

Retrieve relevant evidence first, then answer from that smaller set. A fixed workflow: missed evidence cannot be recovered.

## Early 2025 to mid 2026

The model decides what evidence to collect, how, and when it has enough. Repeated calls make time and cost hard to bound.

Snowflake indicates typically frozen, fire indicates typically trained

**Next** *We can see this kind of behaviour in language model assistants too.*

# *From Research to Products: LLM Assistants*

How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?

Read a file, updated tasks

Added task Locate pages for Figure 4.2 and Table 4.2

Request
Description
Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually

Request
Description Read PDF pages 60-61 and extract the bar heights and token statistics

Added task Verify and answer

Request
Description Cross-check figure values against surrounding text, then answer

Completed task Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

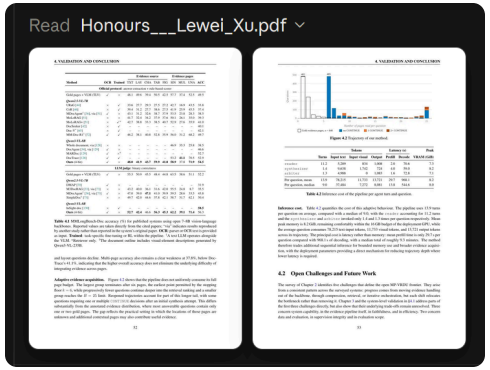
Read Honours___Lewei_Xu.pdf

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

Completed task Verify and answer

- We can see these approaches in assistants: **upload a document and ask questions** about it.
- **All text and page images in the model's context** resembles the second family, but long documents run into context limits.
- **Access through tools**, retrieving or inspecting parts as needed, resembles retrieval-augmented and adaptive-trajectory pipelines.
- Here: a difficult question over my thesis, almost **100 pages**. It plans, locates the pages, and **renders them as images**.




**Next** *Back to the survey: the systems, grouped by family.*

# Surveyed Systems at a Glance

| Model | Venue | OCR | Vision Encoder | LLM Backbone | Mod. | Search | Reasoning | Agentic |
|---|---|---|---|---|---|---|---|---|
| **Page-to-Document Encoding Models** | | | | | | | | |
| Arctic-TILT (2025) | ACL | Yes | U-Net | T5-Large | T, V, L | - | - | - |
| GRAM (2024) | CVPR | Yes | DocFormerv2 | DocFormerv2 | T, V, L | - | - | - |
| Hi-VT5 (2023) | PR | Yes | DiT | T5-Base | T, V, L | - | - | - |
| RM-T5 (2024) | DAS | Yes | DiT | T5-Base | T, V, L | - | - | - |
| **MLLM-Centric Adaptation Models** | | | | | | | | |
| DocR1 (2026) | AAAI | No | Qwen2.5-VL | Qwen2.5-VL-7B | V | - | CoT, HD | - |
| DocSeeker (2025) | CVPR | No | Qwen2.5-VL ViT | Qwen2.5-VL-7B-Instruct | V | - | CoT, HD | - |
| Leopard (2024) | TMLR | No | SigLIP-SO400M | LLaMa-3.1-8B, Mistral-7B | V | - | CoT | - |
| mPLUG-DocOwl2 (2025) | ACL | No | ViT-L/14 | mPLUG-Owl2 | V | - | - | - |
| URaG (2026) | AAAI | No | Qwen2.5-VL ViT | Qwen2.5-VL-3B / Qwen2.5-VL-7B | V | - | HD | - |
| Docopilot (2025b) | CVPR | No* | InternViT-300M | Multiple (Intern) | V | - | - | - |
| Texthawk2 (2024) | preprint | No* | SigLIP-SO400M | Qwen2-7B-Instruct | V | - | - | - |
| DocSLM (2025) | CVPR | Yes | SigLIP2 | Qwen2.5-1.5B | T, V, L | - | - | - |
| InstructDoc (2024) | AAAI | Yes | CLIP | FlanT5, BLIP2 | T, V, L | - | - | - |
| LayTokenLLM (2025b) | CVPR | Yes | - | LLaMA3-8B, Qwen1.5-7B | T, L | - | - | - |
| CoR (2026) | preprint | Yes* | Qwen2.5-VL | Qwen2.5-VL-7B | T, V | - | CoT, PE, HD | - |
| **Retrieval-Augmented Pipelines** | | | | | | | | |
| AVIR (2025) | MMAsia | No | Pix2Struct | Qwen2.5-VL-3B-AWQ | V | D | - | - |
| DFVC (2026) | Electronics | No | VisRAG, ColQwen | Qwen2-VL-2B-Instruct | V | D | - | - |
| DREAM (2025) | MM | No | InternVL2 ViT | InternVL2-4B | V | J | HD | - |
| M3DocRAG (2024) | preprint | No | ColPali, ColQwen | Multiple (Qwen/Intern/Idefics) | V | D | - | - |
| MM-R5 (2026) | AAAI | No | Qwen2.5-VL ViT | Qwen2.5-VL-7B | V | D | CoT | - |
| MoLoRAG (2025b) | EMNLP | No | ColPali | Multiple (Qwen/DS/LLaVA) | V | D, Nav | - | 1A |
| Self-Attention Scoring (2024) | ICDAR | No | Pix2Struct | Pix2Struct | V | D | - | - |
| SLEUTH (2025) | preprint | No | ColPali, Qwen3-VL ViT | Multiple (Qwen/GLM/Gemini) | V | D | HD | 4A |
| SV-RAG (2025) | ICLR | No | InternVL2, Phi-3-vision | Multiple (Intern/Pali/Phi) | V | D | - | - |
| VDocRAG (2025) | CVPR | No* | Phi3V-4.2B | Phi3V-4.2B | V | D | - | - |
| CREAM (2024) | MM | Yes | Pix2Struct | LLaMA2-7B | T, V | J | HD | - |
| KGP (2024c) | AAAI | Yes | - | Multiple | T, L | S, IR, Nav | - | 1A |
| MHier-RAG (2025) | preprint | Yes | Qwen-VL-Plus | Multiple (Qwen/DS/GPT) | T, V, L | J | CoT, HD | - |
| MLDocRAG (2026) | preprint | Yes | Qwen2.5-VL | Multiple (Qwen) | T, V, L | D | CoT, HD | - |
| MultiDocFusion (2025) | ACL | Yes | - | Multiple (Qwen/LLaMa/Mistral) | T, V, L | D | HD | - |
| M2RAG (2025a) | NeurIPS | Yes* | VisRAG-Ret | Multiple (Qwen) | T, V | J | - | 3A |
| MDocAgent (2025) | preprint | Yes* | ColPali, ColQwen2 | Multiple (Llama/Qwen) | T, V | J | - | 5A |
| PDF-WuKong (2024) | preprint | Yes* | IXC2-VL-4KHD | IXC2-VL-4KHD | T, V | J | - | - |
| RAG-DocVQA (2025) | ICDAR | Yes* | DiT, Pix2Struct | Qwen2.5-VL-7B-Instruct | T, V, L | D | - | - |
| VisDoMRAG (2025) | NAACL | Yes* | ColPali / ColQwen2 | Multiple(Qwen/Gemini/GPT) | T, V | J | CoT | 3A, SC |
| **Adaptive-Trajectory Pipelines** | | | | | | | | |
| Doc-V* (2026) | preprint | No | Qwen2.5-VL | Qwen2.5-VL-7B-Instruct | V | D, IR, QR, Nav | ReAct, HD | 1A |
| DocReact (2025a) | ACL | No | ColPali, VisRAG | GPT-4o | V | D, IR, QR | ReAct, QD | 1A |
| MACT (2025) | preprint | No | Multiple | Multiple (Qwen/MiMo/Intern) | V | D | CoT, PE | 4A, SC, SV |
| VRAG-RL (2026) | NeurIPS | No | Qwen2.5-VL ViT | Qwen2.5-VL-3B/7B-Instruct | V | D, IR, QR | ReAct, HD | 1A |
| DocLens (2025a) | preprint | Yes | – | Multiple (GPT/Gemini/Claude) | T, V, L | J | CoT | 4A, SA |
| DMAP (2026) | WWW | Yes* | ColPali | GPT-4o / Qwen-plus | T, V, L | J | HD | 2A, SR |
| DocAgent (2025) | EMNLP | Yes* | – | Multiple (GPT/Gemini/Claude) | T, V, L | J, IR, Nav | ReAct, HD | 3A |
| DocDancer (2026) | preprint | Yes* | Qwen3-VL-235B | Qwen3-4B/30B-A3B | T, V, L | J, IR, QR | ReAct | 1A |
| MM-Doc-R1 (2026) | preprint | Yes* | Qwen2.5-VL ViT | Qwen3-4B/8B | T, V | S, IR, QR, Nav | ReAct, HD | 3A |
| SimpleDoc (2025) | EMNLP | Yes* | ColQwen-2.5 | Multiple (Qwen) | T, V | J, IR, QR | - | 1A, SR, SV |
| ViDoRAG (2025a) | EMNLP | Yes* | NV-Embed-V2, ColQwen2 | Multiple (Qwen/LLaMa/GPT) | T, V | J, IR | ReAct, HD | 3A, SR, SV |

1

## Models and pipelines

The first two families are **models**: generally trained end to end, with components built for that architecture.

The last two are **pipelines**: components are often interchangeable, **especially the backbone**. As language models get stronger, the pipelines benefit too.

2

## Modalities

Some systems use **vision only**, reading page images. Others combine images with **text and layout**: text embedded in the file or extracted by a parser, and layout as bounding boxes or coordinates.

This distinction matters in the empirical study.

**Next** *That brings us to the second paper.*

# *Table of Contents*


- Introduction
- Survey of MP-VRDU
- **Empirical Attribution Study**

  *Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution*

  Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

  **Under review**
- Conclusion
- Proof of Concept: A Simple Method


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# *Real-World Constraints*

Assistants are one application. In industry: one or a few specific tasks, over **thousands to millions of documents**.

**Healthcare**
Extract fields from scanned patient records and referrals.

**Mining and resources**
Classify and summarise technical and safety reports.

**Finance**
Pull figures from annual reports and filings.

**Legal**
Find the relevant clauses across contracts.

**Privacy**
Privacy or data-residency requirements may rule out a third-party service.

**Cost**
Even without privacy concerns, a commercial API is very expensive at this volume.

**Scale**
Uploading documents **one by one** to an assistant is not feasible. You may also need **structured output** for another system.

API or local model, you still have to **implement and deploy your own system**, tailored to your task, documents and output. API access does not give you the workflow behind the consumer assistant.

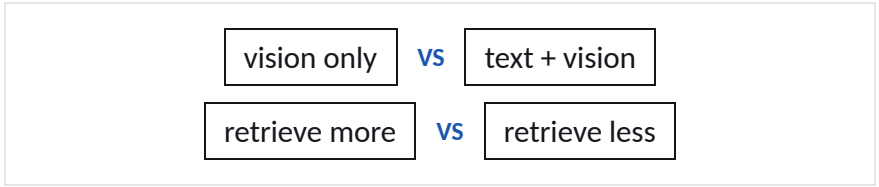
Convert ▸ Retrieve ▸ Use evidence

These design and deployment choices, **within a fixed compute budget**, motivate our empirical study.

# *Empirical Study: Research Gaps and Contributions*

## RESEARCH GAPS

1 **Competing design claims**, each tested on a different pipeline and dataset.




2 **No end-to-end empirical account** of where failures occur across a document understanding pipeline.

## CONTRIBUTIONS

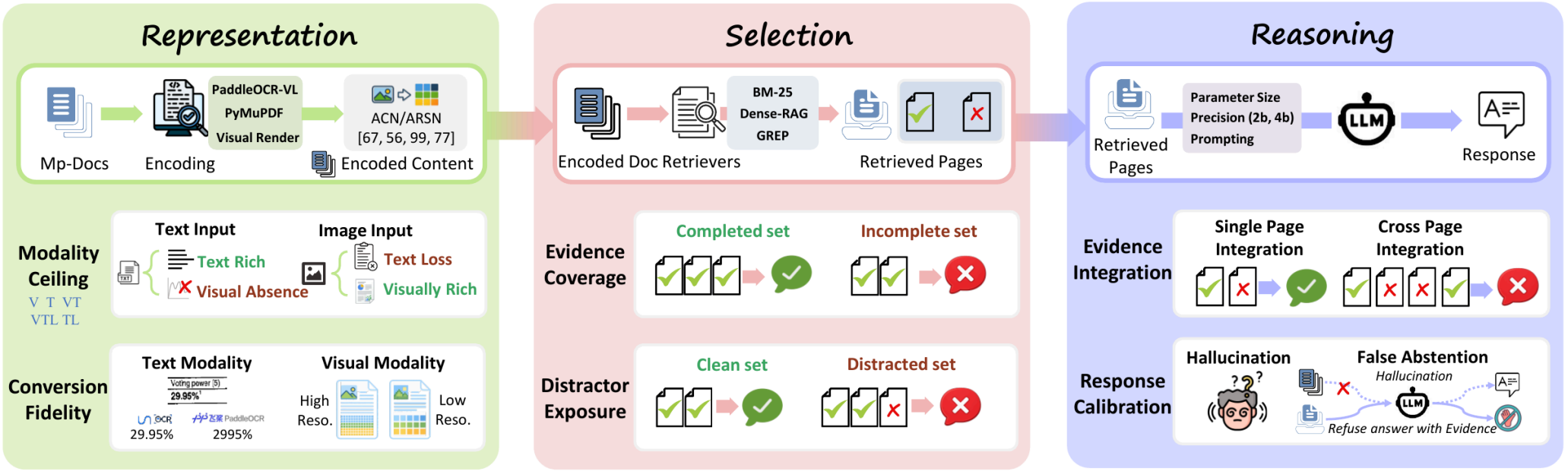
1 A **unified view of failure**: three places an answer can go wrong, regardless of the architecture.

2 Controlled interventions in **one pipeline**, changing one stage at a time, repeated across document collections and independently developed model families.

3 **Concrete guidance** for designing and deploying systems **under a compute budget**.

**Next** *First, the framework: three places an answer can go wrong.*

# Attribution Framework




**REPRESENTATION**

Was the information preserved when the page was converted?

**SELECTION**

Did it actually reach the model?

**REASONING**

Did the model use it correctly?

---

Each locus has **two mechanisms**, six in total. Every technique targets one or more of these loci, whatever the architecture.

# *Attribution by Construction*

One simple pipeline. **Change one stage, hold the other two fixed.**




- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.
- **Two model families** (Qwen3-VL and Gemma-3) and a second collection (LongDocURL) check that the findings transfer.

**Next** *We ran many experiments. The next three slides cover only the most interesting and important points.*

# *Representation: When Is Visual Reading Worth Its Cost?*

### 1 Vision is expensive

At the same 900 input tokens, page images take **21× longer** to read than text (22.7 s against 1.1 s a question) and run out of memory on 11% of questions, against 1%.

### 2 Yet most systems read everything visually

Vision only, or text plus image, for every page. But the best choice depends on the **document type** and the **task**: scans have no text layer; born-digital text and tables gain little from the image; charts and figures need it.

### 3 Modalities are complementary

Text alone 38%, a low-resolution image alone 38%, together 49%. They fail on **different questions** (only 22% are answered by both), so parser noise and low resolution cover for each other.

### 4 Deployment: choose, don't default

A known document type or task fixes the modality: images for scans, cached parser text for reports and tables. Mixed collections: choose per question, as tool-using agents are starting to.

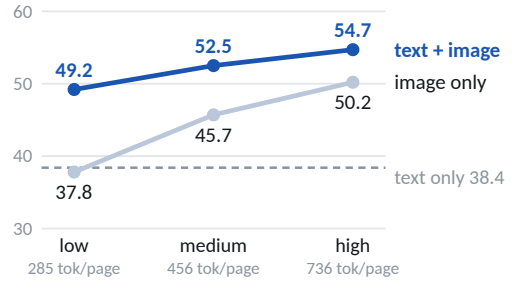
**By evidence source and representation**

| | Abstention (%) | | | | Accuracy (%) | | | |
|---|---|---|---|---|---|---|---|---|
| Evidence | T | TL | TLV | V | T | TL | TLV | V |
| **MMLongBench-Doc, scanned documents** | | | | | | | | |
| Text | 70.8 | 18.5 | 12.3 | 12.3 | 3.1 | 35.4 | **49.2** | **49.2** |
| Layout | 52.6 | 18.4 | 0.0 | 2.6 | 2.6 | 23.7 | **55.3** | 47.4 |
| Table | 81.2 | 12.5 | 6.2 | 9.4 | 0.0 | 59.4 | **68.8** | 59.4 |
| Chart | 77.1 | 43.8 | 16.7 | 12.5 | 0.0 | 4.2 | **52.1** | 50.0 |
| Figure | 61.3 | 19.8 | 4.7 | 3.8 | 6.6 | 31.1 | **52.8** | 50.0 |
| *All* | 69.2 | 21.9 | 6.5 | 6.0 | 3.5 | 32.3 | **57.7** | 55.2 |
| **LongDocURL, born-digital** | | | | | | | | |
| Text | 4.6 | 3.5 | 1.5 | 1.6 | 71.8 | 75.6 | **78.2** | 70.0 |
| Layout | 2.1 | 1.6 | 0.7 | 0.9 | 45.0 | 49.2 | **51.3** | 49.3 |
| Table | 5.0 | 3.2 | 1.1 | 0.8 | 57.7 | **65.6** | 63.6 | 52.4 |
| Figure | 6.7 | 8.6 | 1.5 | 1.3 | 59.9 | 62.8 | **69.9** | 61.9 |
| *All* | 4.7 | 4.3 | 1.3 | 1.3 | 60.3 | 64.5 | **67.3** | 59.4 |

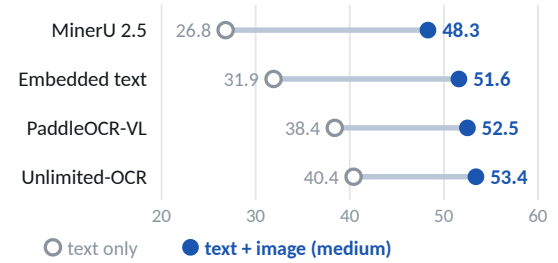
*T embedded text, TL parsed text, TLV parsed text plus page image, V page image.*

**Each modality hides the other's weakness**

**Text covers for low resolution**




**The image covers for parser noise**




> ***No one size fits all: choose modalities for the document and the task, and combine cheap ones before paying for resolution.***

**Next** *Once the pages are readable, which ones should the model see?*

# *Selection: How Much Evidence Is Enough?*

1. **Evidence can look sufficient**

   The model produces an answer either way. It is hard to know whether more context from the document, such as an exception or a qualification on another page, would change it.

2. **Too little: evidence is missing**

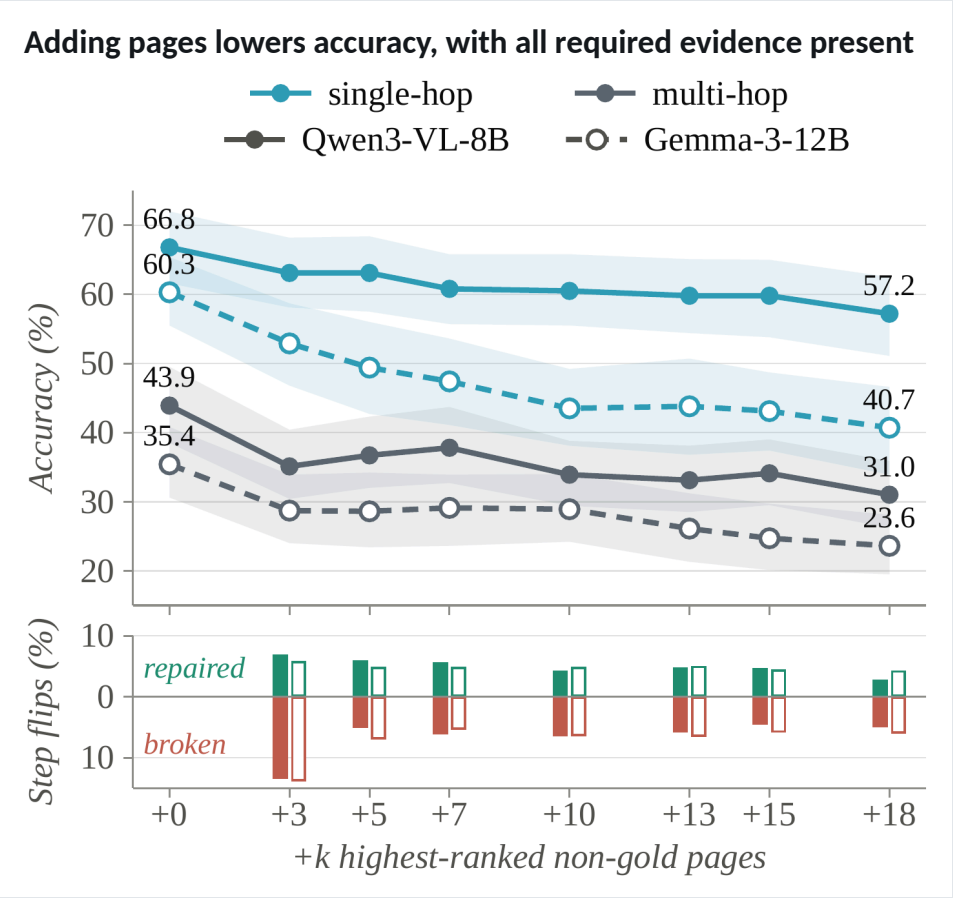
   Removing one required page halves accuracy, from 38.6% to 18.4%, and retrieval rank does not show which page is safe to drop.

3. **Too much: evidence gets buried**

   With every required page present, extra pages still lower accuracy. The reasoner struggles to find and integrate what matters, most of all on multi-hop questions.

4. **No clear stopping signal**

   It is hard to retrieve only what is relevant, and hard to know when to stop looking for more.




> ***We need enough evidence to establish the answer, not just enough to generate one.***

# *Reasoning: Does the Document Support the Question?*

## 1 Unsupported premises

A model answers fluently even when the document does not support the question. With no instruction it refuses only **45%** of unanswerable questions.

> "Which YouTube does the slides use…?" The document mentions no video.
> ✗ "A clip from *Breaking Bad*…" ✓ "Not answerable" once it may abstain.

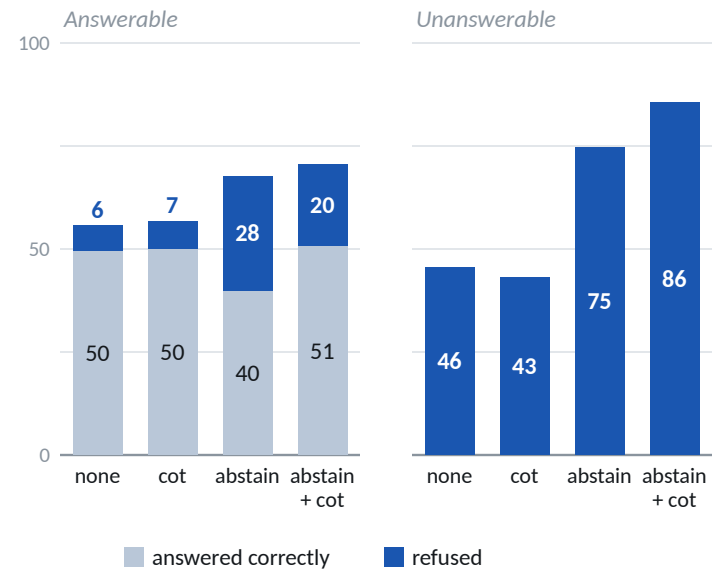
## 2 Prompts move the trade-off

Permitting abstention lifts refusal to **75%**, but also refuses 28% of answerable questions. Reasoning first, then abstaining, wins the accuracy back (51%) and refuses more of the unanswerable (86%).

## 3 More computation helps, at a price

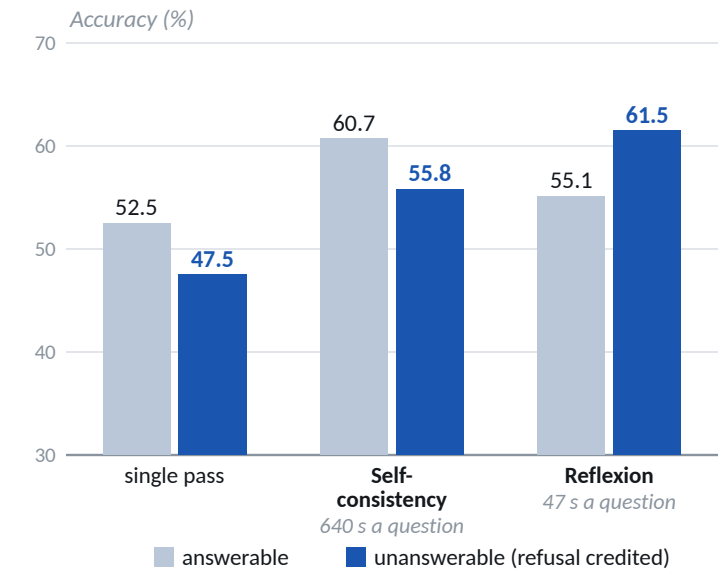
**Self-consistency** (40 sampled reasoning paths, majority vote) adds 8 points on answerable questions, at 640 s a question. **Reflexion** (retry on its own evaluation) adds 14 on unanswerable ones at 47 s, though its self-check passes 94% of its answers.

**Prompting: answering and refusing**




*Qwen3-VL-8B with gold pages, share of questions (%).*

**Spending more computation on reasoning**




*Self-consistency on 773 of 847 answerable questions so far; single pass is the uninstructed prompt.*

> ***Before asking whether the answer is correct, ask whether the evidence supports giving that answer.***

# *Conclusion*

SURVEY

## Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

## Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

# Thank you

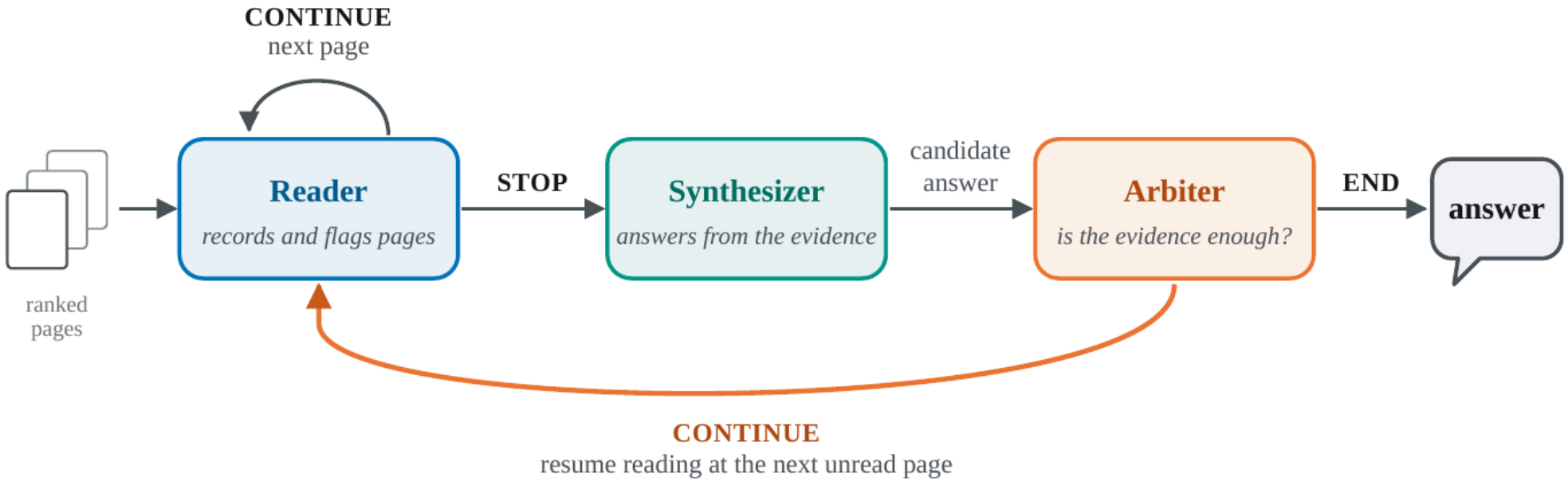
Questions?

# *Table of Contents*


- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- **Proof of Concept: A Simple Method**


THE UNIVERSITY OF WESTERN AUSTRALIA

# *From Findings to a Simple Method*




**REPRESENTATION**

Every page is read as **text and image together**.

**SELECTION**

**Read broadly**, then pass on compact records, not full pages.

**REASONING**

A separate arbiter decides **whether the evidence is enough**.

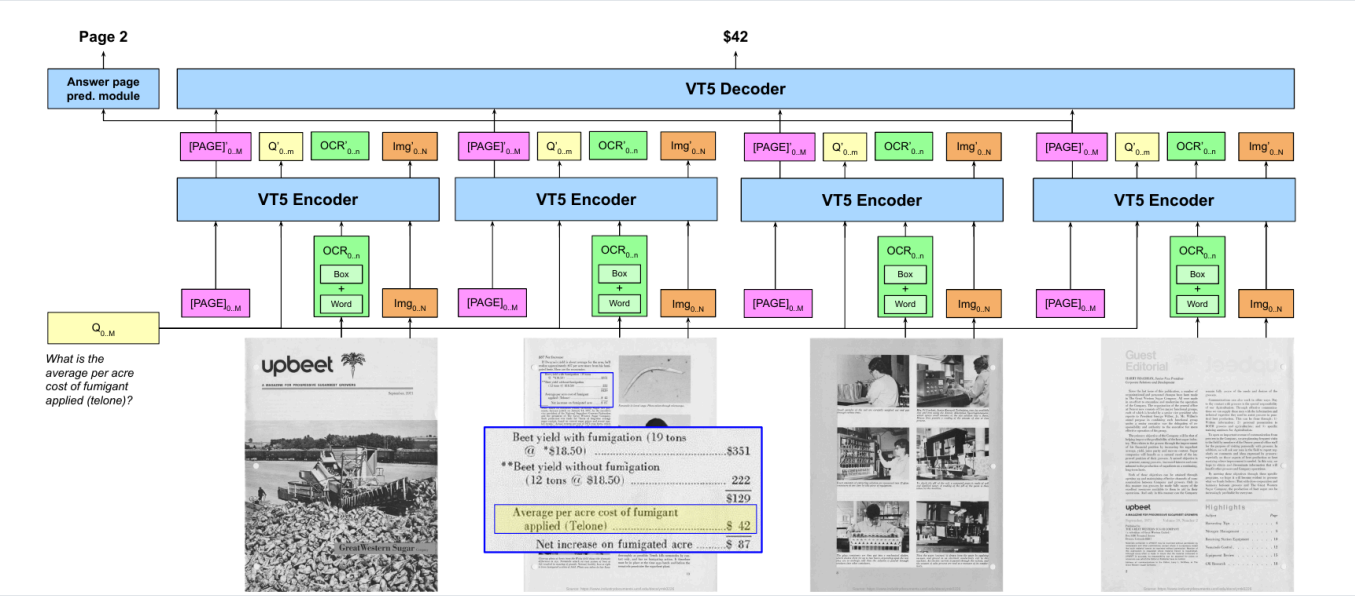
# *Simple Method Results*

| Method | OCR | Trained | MMLongBench-Doc | | | | LongDocURL | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | SIN | MUL | UNA | **ACC** | UND | REA | LOC | **ACC** |
| Gold pages + VLM | ✓ | × | 57.7 | 37.4 | 52.5 | 49.5 | 69.7 | 63.6 | 59.5 | 65.7 |
| Whole document (via DocTrace) | × | × | 46.9 | 35.3 | 25.8 | 38.5 | – | – | – | 45.1 |
| M3DocRAG (via SLEUTH) | × | × | – | – | 48.6 | 45.9[a] | 60.1 | 52.4 | 45.5 | 54.6 |
| DocAgent (via MARDoc) | ✓ | × | – | – | – | 46.6 | – | – | – | – |
| MDocAgent (via SLEUTH) | ✓ | × | – | – | 49.7 | 47.8[a] | 57.5 | 51.5 | 44.7 | 53.1 |
| MoLoRAG (via SLEUTH) | × | × | – | – | 51.7 | 48.8[a] | 63.4 | 52.7 | 49.3 | 57.6 |
| SLEUTH | × | × | – | – | 67.4 | 52.8[a] | 65.7 | 53.0 | **53.6** | 60.0 |
| DocTrace | ✓ | ✓ | 53.2 | **41.1** | 70.5 | 52.9 | – | – | – | 56.4 |
| **Ours** (4-bit) | ✓ | × | **60.5** | 39.2 | **76.9** | **56.5** | **67.2** | **66.3** | 47.9 | **61.3** |

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

# *Page-to-Document Encoding: Hi-VT5*

**Page-to-document encoding** | MLLM-centric adaptation | Retrieval-augmented | Adaptive-trajectory



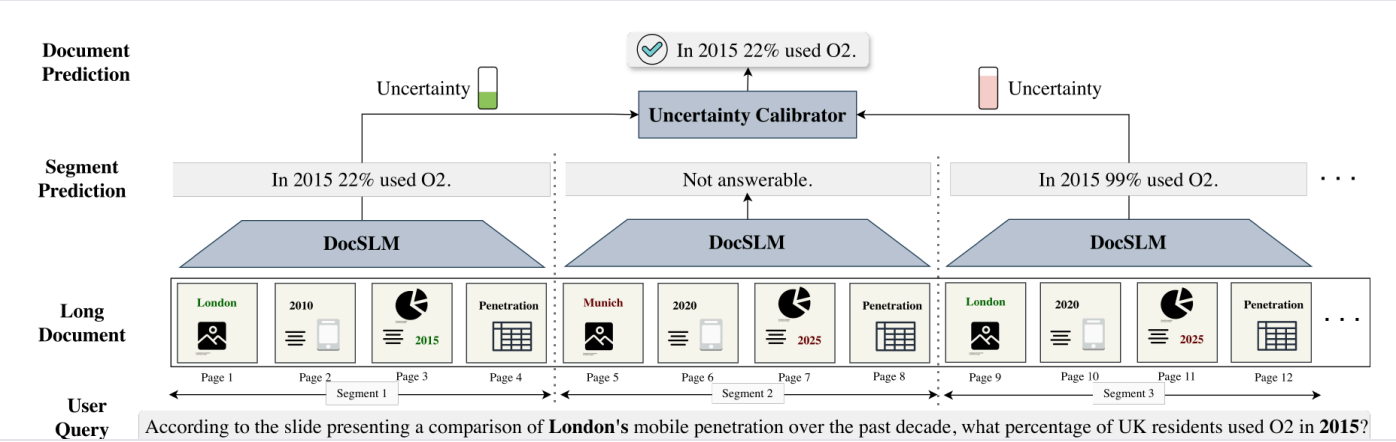

*Hi-VT5 architecture. Each page is encoded into [PAGE] tokens, and the decoder reads the tokens of all pages to generate the answer. A second module predicts the answer page. From Tito et al., Pattern Recognition 2023.*

**HOW IT WORKS**

- Builds on **T5**, a text encoder-decoder, and adds visual and layout information.
- Each page is **encoded separately**, with the question, the recognised words and their positions, and patches of the page image.
- Ten learnable **[PAGE] tokens** summarise what on that page is relevant to the question.
- The decoder reads the **joined page tokens**, not the full pages, and generates the answer.

# *MLLM-Centric Adaptation: DocSLM*

Page-to-document encoding | **MLLM-centric adaptation** | Retrieval-augmented | Adaptive-trajectory



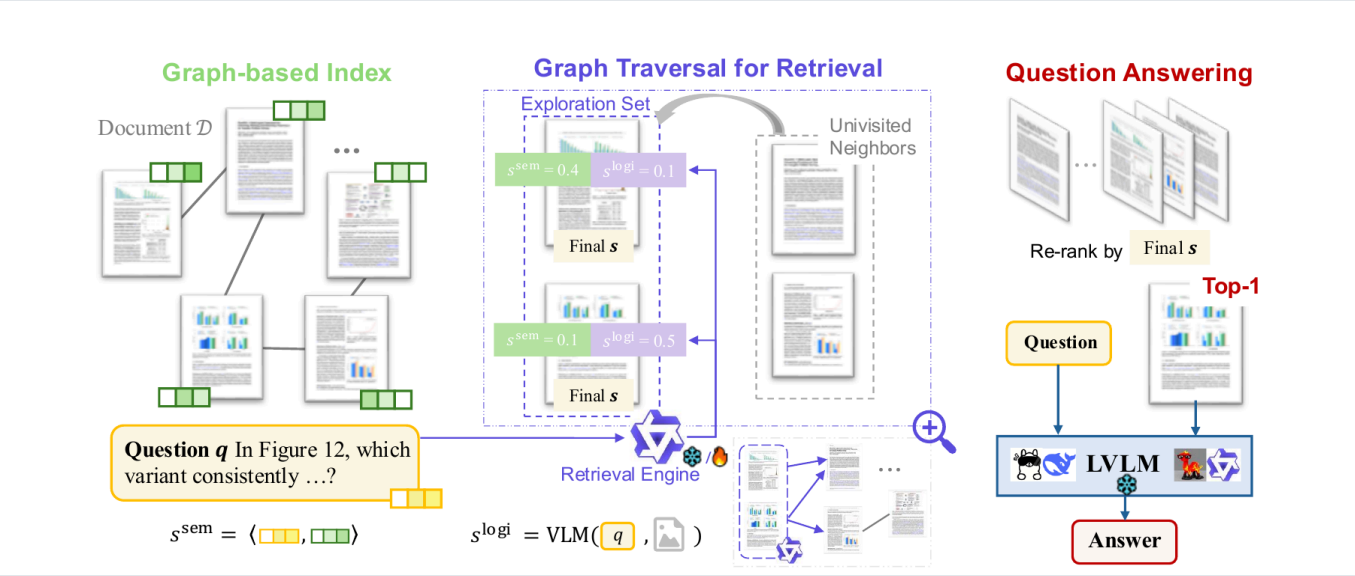

*Streaming abstention in DocSLM. The document is split into segments. Each segment gives an answer or abstains, and an uncertainty calibrator keeps the answer with the lowest uncertainty. From Hannan et al., CVPR 2026 Findings.*

**HOW IT WORKS**

- A **small vision-language model** for long documents **under limited memory**.
- **Hierarchical multimodal compressor.** Each page's visual, text and layout information is jointly encoded into a **fixed 576 tokens**.
- **Streaming abstention.** The document is split into segments and processed one at a time. Each segment answers or abstains.
- An **uncertainty calibrator** keeps the answer with the lowest uncertainty.

# *Retrieval-Augmented: MoLoRAG*

Page-to-document encoding | MLLM-centric adaptation | **Retrieval-augmented** | Adaptive-trajectory



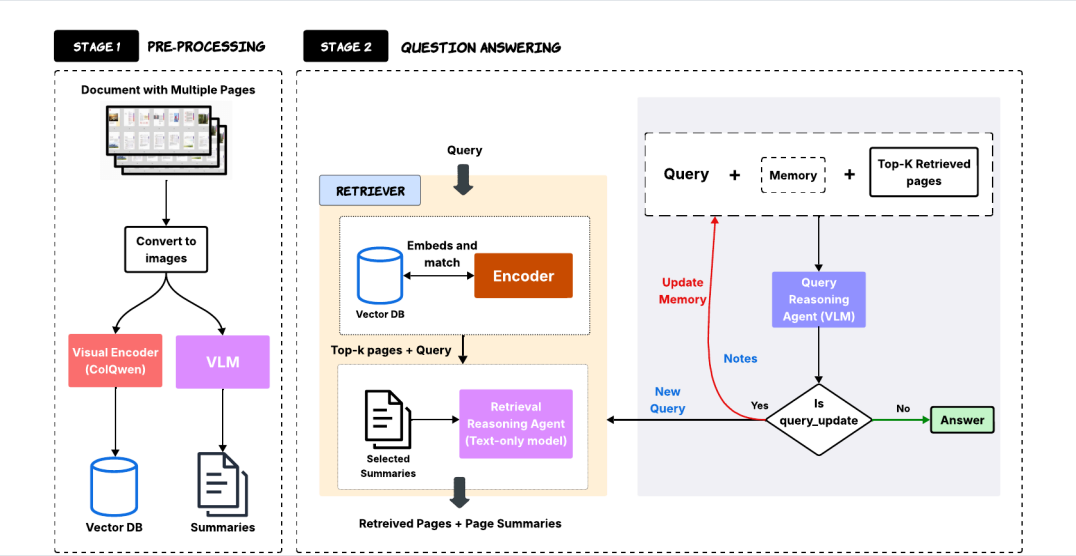

*The MoLoRAG framework: a graph-based index over pages, graph traversal for retrieval using semantic and logical relevance scores, then question answering over the top pages. From Wu et al., EMNLP 2025.*

HOW IT WORKS

- **Graph-based index.** A visual retriever represents each page, and pages are connected by their similarity.
- **Graph traversal.** The search starts from pages similar to the question. A small vision-language model helps score relevance, and the top K pages are kept.
- **Question answering.** The top K pages go to a separate vision-language model, the reasoner.

# *Adaptive-Trajectory: SimpleDoc*

Page-to-document encoding | MLLM-centric adaptation | Retrieval-augmented | **Adaptive-trajectory**




*SimpleDoc. Stage 1 indexes each page as an embedding and a summary. Stage 2 retrieves and re-ranks pages, and a reasoning agent answers or updates its memory and query. From Jain et al., EMNLP 2025.*

HOW IT WORKS

- **Pre-processing.** Every page is indexed twice: as a **visual embedding**, and as a short **summary** written by a vision-language model.
- **Dual-cue retrieval.** Pages are shortlisted by embedding similarity, then a language model **re-ranks them using the summaries**.
- **Reasoning agent.** It reads the retrieved pages with a **working memory** of earlier notes, then either answers or **issues a refined query** to retrieve more.
- The loop ends when it answers, decides the question is not answerable, or reaches a step limit.

# *Inference-Time and Training Strategies*

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

## Retrieval and navigation

**TRAINING-FREE**

Similarity-based or relation-aware retrieval

**TRAINED**

Modality-alignment, cross-page-fusion, logical-relevance

## Reasoning strategies

**TRAINING-FREE**

Chain-of-Thought, Self-Reflection, Self-Consistency

**TRAINED**

Trace-imitation, then reward-driven (GRPO)

## Agentic methods

**TRAINING-FREE**

ReAct and tool use, multi-agent orchestration

**TRAINED**

Trajectory-distillation, reinforcement-learned

# *Retrieval and Navigation*

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

# *Reasoning Strategies*

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

# *Agentic Methods*

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

# *Training Strategies*

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

## Trained retrieval

**Modality-alignment** aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

## Trained reasoning

**Trace-imitation** supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

## Trained agentic methods

**Trajectory-distillation** clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

# *Datasets*

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

## Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

## Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

## Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

**Answer-only accuracy masks retrieval failure** and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

# *Datasets and Evaluation*

PRIMARY

## MMLongBench-Doc

| 135 | 48.3 | 847 | 244 |
|---|---|---|---|
| documents | mean pages | answerable | unanswerable |

- Annotated with **evidence pages, evidence sources and answerability**, which the experiments need.
- 367 of the answerable questions need two or more pages.

COMPLEMENTARY

## LongDocURL

| 396 | 85.6 | 2,325 |
|---|---|---|
| documents | mean pages | questions |

- **Born-digital only**, so answers are often in the text layer, even for figure questions.
- Used for the modality-ceiling result only: its evidence pages are not a minimal set.

**Judge:** Gemini-2.5-Flash at temperature zero returns correct, incorrect or abstained. Agreement is **96.0%** with GPT-4o and **98.3%** with a human annotator.

# *Modality Ceiling*

| | Abstention | | | | Accuracy | | | |
|---|---|---|---|---|---|---|---|---|
| **Evidence** | **T** | **TL** | **TLV** | **V** | **T** | **TL** | **TLV** | **V** |
| **MMLongBench-Doc** | | | | | | | | |
| Text | 23.0 | 11.5 | 7.3 | 8.0 | 36.9 | 44.4 | **51.0** | 42.3 |
| Layout | 20.3 | 10.2 | 0.8 | 4.2 | 26.3 | 33.9 | **49.2** | 43.2 |
| Table | 15.9 | 7.2 | 6.7 | **9.1** | 35.6 | **44.7** | **44.7** | 34.1 |
| Chart | 32.6 | **28.1** | 7.3 | 6.2 | 16.3 | 16.9 | **42.7** | 40.4 |
| Figure | **36.6** | 24.8 | **7.6** | 7.9 | 14.5 | 22.1 | **42.4** | 39.0 |
| *All* | 26.3 | 17.2 | 6.2 | 7.1 | 29.0 | 35.7 | **49.5** | 42.6 |
| **LongDocURL** | | | | | | | | |
| Text | 4.6 | 3.5 | **1.5** | **1.6** | 71.8 | 75.6 | **78.2** | 70.0 |
| Layout | 2.1 | 1.6 | 0.7 | 0.9 | 45.0 | 49.2 | **51.3** | 49.3 |
| Table | 5.0 | 3.2 | 1.1 | 0.8 | 57.7 | **65.6** | 63.6 | 52.4 |
| Figure | **6.7** | **8.6** | **1.5** | 1.3 | 59.9 | 62.8 | **69.9** | 61.9 |
| *All* | 4.7 | 4.3 | 1.3 | 1.3 | 60.3 | 64.5 | **67.3** | 59.4 |

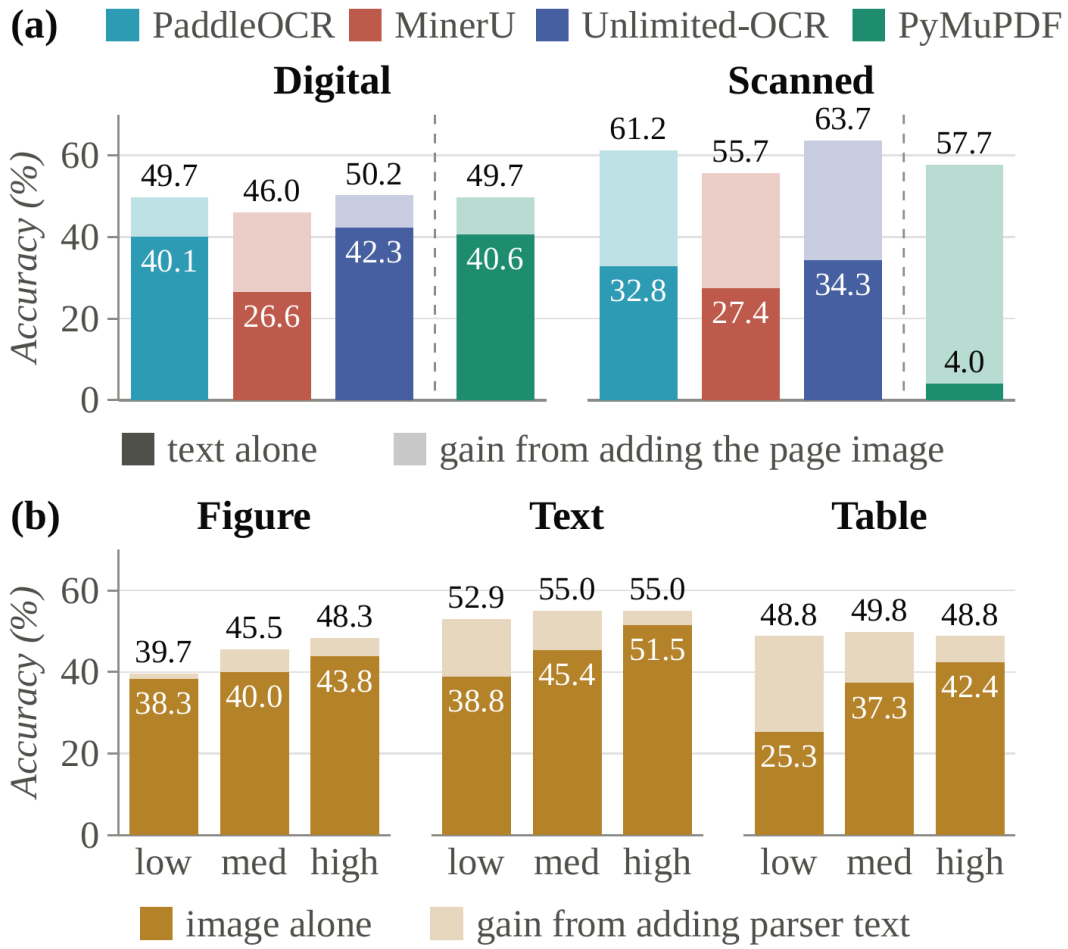



REPRESENTATION

## Evidence accessibility depends on modality

- On MMLongBench-Doc, text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**.
- Images alone stay below text plus images: 42.6% against 49.5%.
- On born-digital LongDocURL, adding vision to parser text gives only **2.8 points** overall, 7.1 on figure questions.

# Conversion Fidelity




**Encode**
parser, resolution

**Retrieve**
gold pages

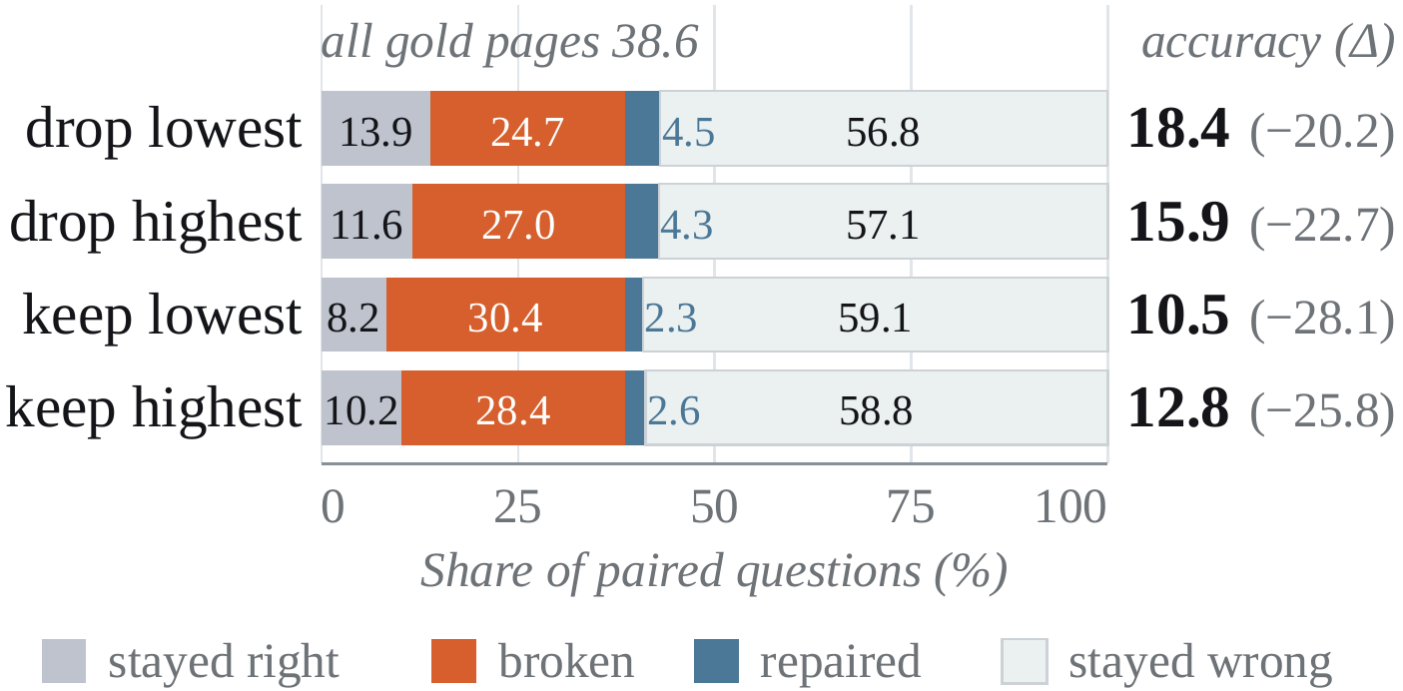
**Answer**
fixed

REPRESENTATION

## Complementary modalities reduce conversion sensitivity

- Adding page images cuts the spread across parsers from **15.7 to 4.2 points** on digital documents.
- Higher resolution helps images alone far more than text plus images.
- Scanned documents stay sensitive to the parser, and figure questions to resolution.

# Evidence Coverage





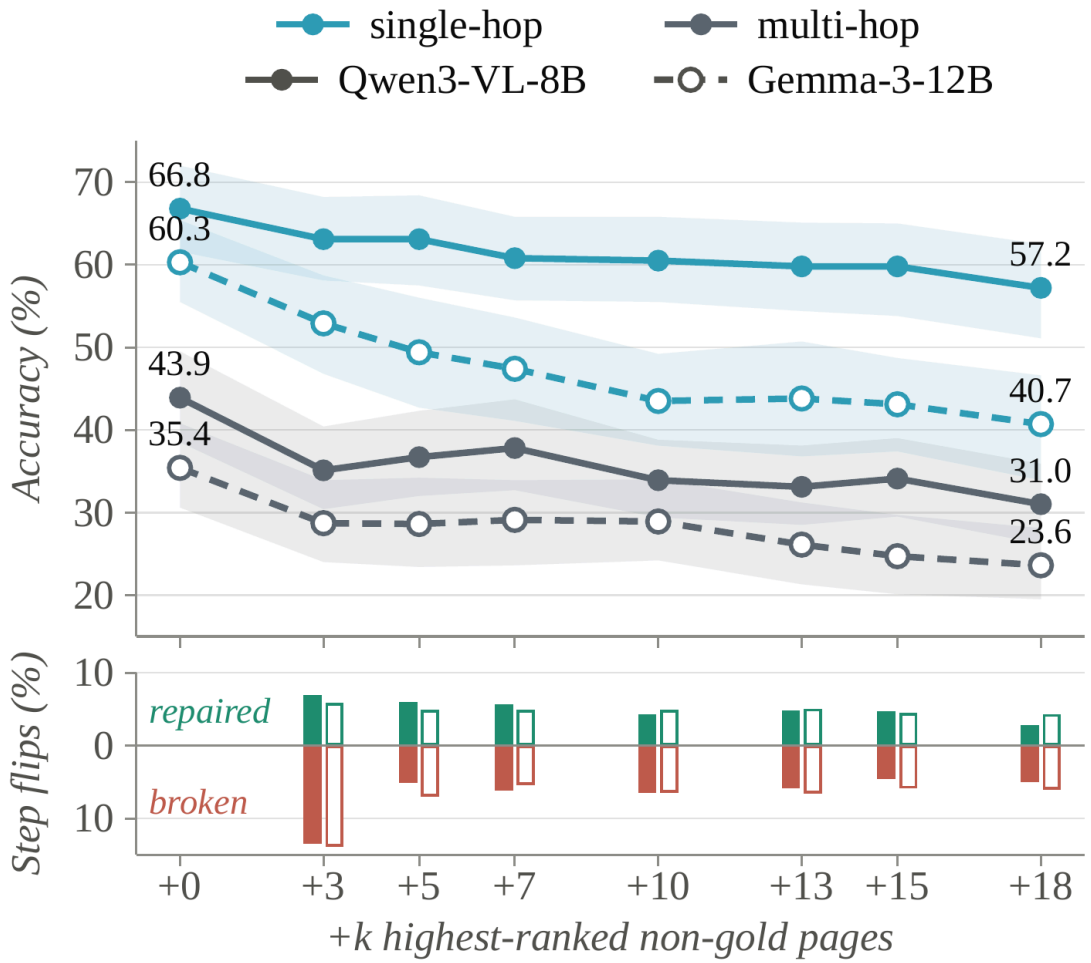


SELECTION

## Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** (lowest-ranked) or 15.9% (highest-ranked).
- Correct answers break far more often than wrong ones are repaired.
- Retrieval rank does not predict which page is safe to lose.

# Distractor Exposure




**Encode**
fixed

**Retrieve**
gold + k other pages

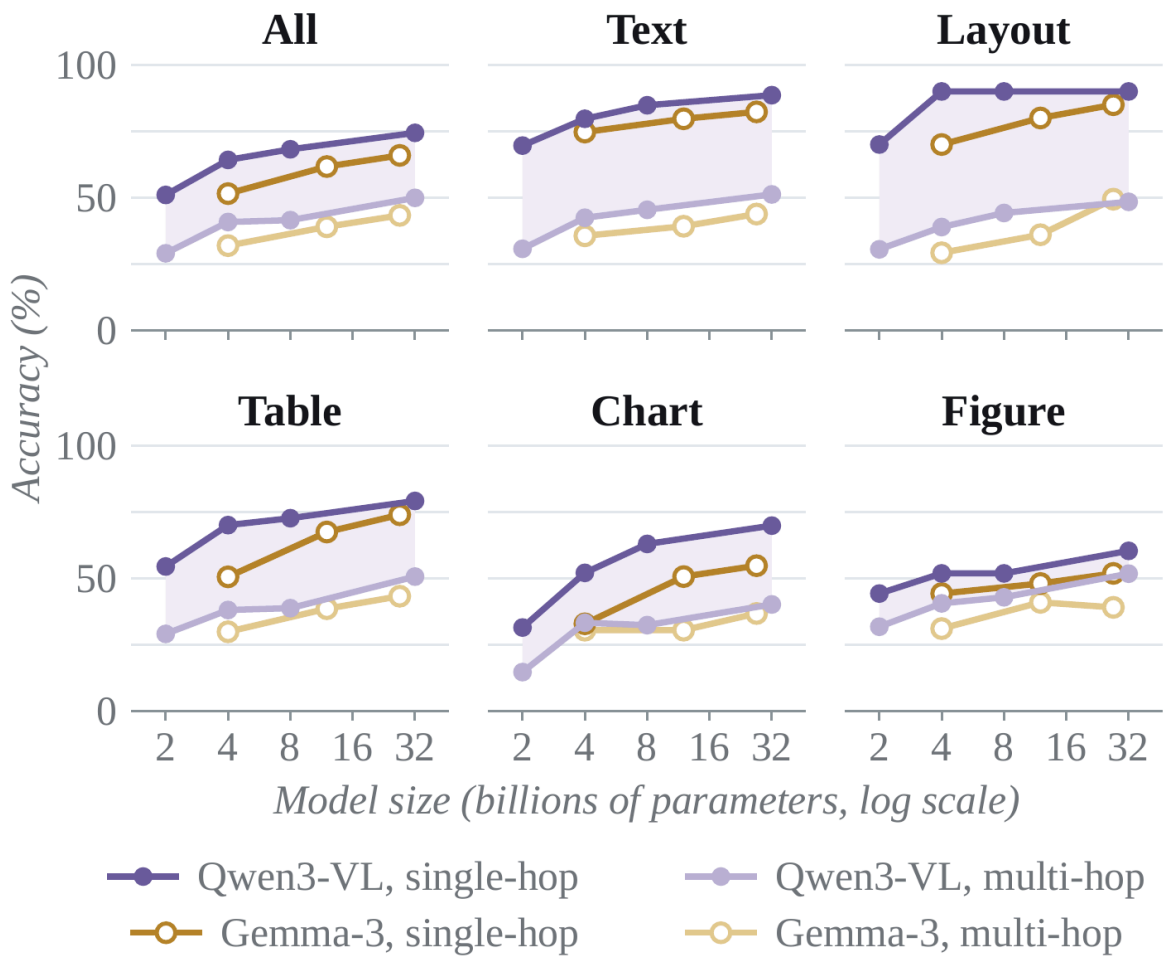
**Answer**
fixed

SELECTION

## Multi-hop accuracy drops sharply with added context

- Qwen3-VL-8B: multi-hop falls from **43.9% to 35.1%** after three extra pages. Single-hop declines gradually, 66.8% to 57.2%.
- Gemma-3-12B shows the same trend: multi-hop 35.4% to 23.6%, single-hop 60.3% to 40.7%.
- Every gold page is still there. Extra context both breaks and repairs answers.

# Evidence Integration




**Encode**
fixed

**Retrieve**
gold pages

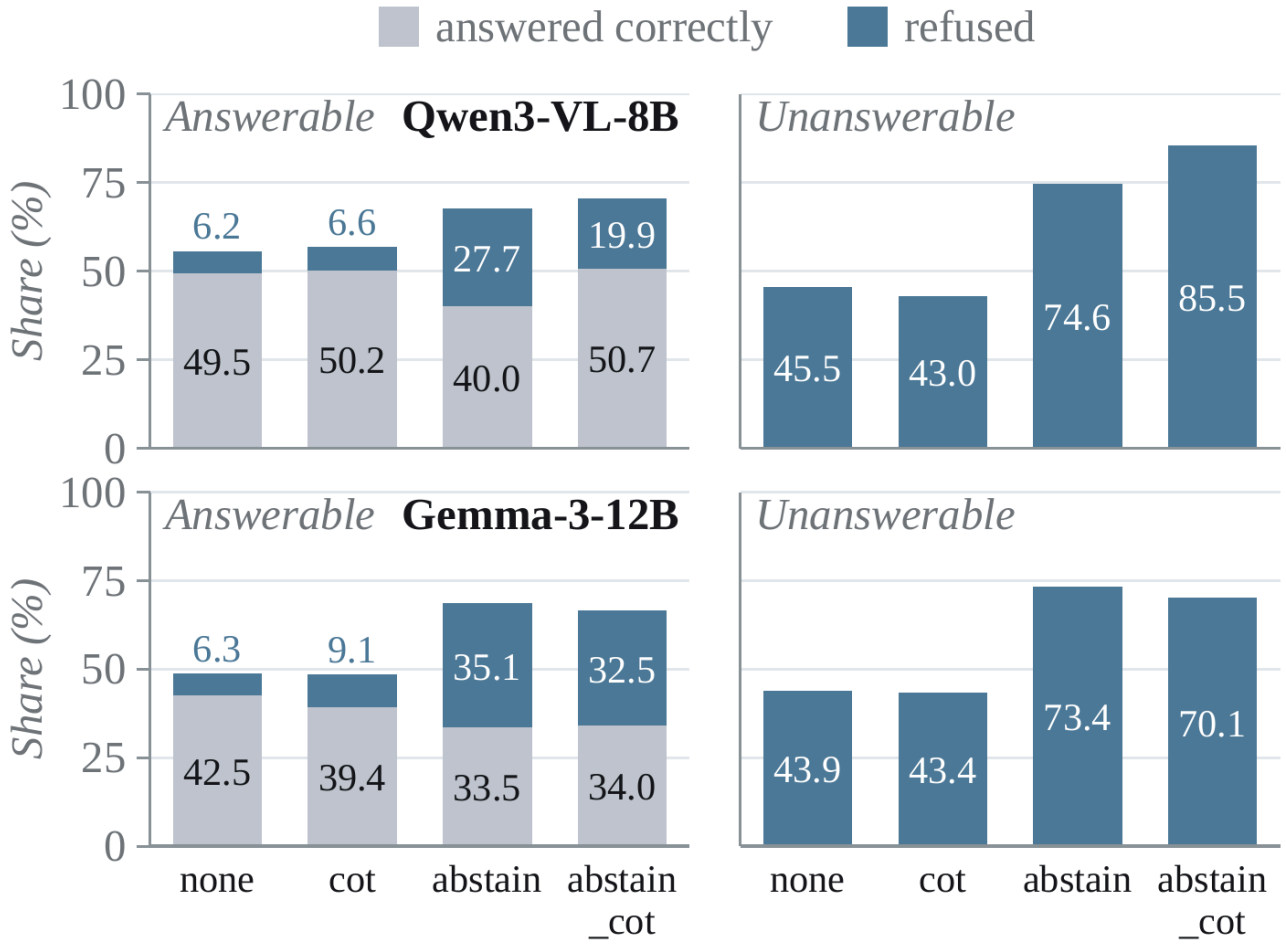
**Answer**
model size

REASONING

## Scaling helps, but the multi-hop gap persists

- Qwen3-VL from 2B to 32B: multi-hop rises from **29.0% to 49.9%**, single-hop from 51.0% to 74.4%.
- The gap between them stays roughly the same.
- Gemma-3 shows the same separation.

# Response Calibration




**Encode**
fixed

**Retrieve**
gold pages

**Answer**
prompt mode

REASONING

## Reasoning before abstaining limits false refusal

- Qwen3-VL-8B, abstain: unanswerable refusal **45.5% to 74.6%**, but false refusal **6.2% to 27.7%** and accuracy 49.5% to 40.0%.
- Adding CoT: false refusal 19.9%, accuracy 50.7%, unanswerable refusal 85.5%.
- Gemma-3-12B gains much less from CoT: false refusal 35.1% to 32.5%.

# *Representation: Design and Deployment*

*Is the evidence in the input?*

## Modality ceiling

*two inline icons, parsed as <img>*

**Q** What are the colors of the icons that the users touch to move the app up and remove the app respectively?

**Gold** gray, red

| | | |
|---|---|---|
| **T** | ✗ | "The document does not specify the colors" |
| **TL** | ✗ | "it does not mention any visual details" |

▼ **add the page image**

| | | |
|---|---|---|
| **V** | ✓ | "The icon is gray. … The icon is red." |
| **TLV** | ✓ | "gray (a gray upward arrow) … red" |

*Colour exists only in pixels; a text input leaves the reasoner nothing to read.*

## Conversion fidelity

*scanned slide, text by parser*

**Q** What is the name of the company that created this presentation?

**Gold** AVALAUNCH MEDIA

| | | |
|---|---|---|
| **PaddleOCR** | ✗ | "Asterks" (parse: KICK ASTERKS) |
| **MinerU** | ✗ | "does not provide any evidence" |

▼ **re-convert, or add the image**

| | | |
|---|---|---|
| **Unlimited** | ✓ | "AVALAUNCH MEDIA" |
| **with image** | ✓ | "Avalaunch Media … in the banner" |

*The error sits in the transcript, not the reasoner, and each parser fails differently.*

**DESIGN** — **Complementary modalities** recover what one input loses: prioritise reliable embedded text, add images when appearance matters or extraction is unreliable, re-parse when the text is inadequate.

**DEPLOYMENT** — Embedded text plus images matches parser text plus images (**50.1% against 49.5%**) with fewer tokens. Images cost **26.3 s** of prefill against 1.8 s, so set modality and selection depth together.

# *Selection: Design and Deployment*

*Is the right evidence in context?*

## Evidence coverage

*author list on pages 13, 14*

**Q** How many chapter authors of this report come from Columbia University?

**Gold** 3

| | | |
|---|---|---|
| **page 13** | ✗ | “two” (Horton, Kinney) |
| **page 14** | ✗ | “one” (Garcia-Pando) |

▼ **add the complementary page**

| | | |
|---|---|---|
| **both pages** | ✓ | “three” (Horton, Kinney, Garcia-Pando) |

*Each page is relevant, yet neither suffices: the answer is a property of the page set.*

## Distractor exposure

*10 gold pages of a 20-page deck*

**Q** How many lateral bar graphs are present in the document?

**Gold** 10

| | | |
|---|---|---|
| **+3 pages** | ✗ | “there are 12 lateral bar graphs” |
| **+5 pages** | ✗ | “there are 13 lateral bar graphs” |

▼ **remove competing pages, or read on**

| | | |
|---|---|---|
| **gold pages** | ✓ | “there are 10 lateral bar graphs” |
| **+7 to +18** | ✓ | “there are 10 lateral bar graphs” |

*A few added pages inflate the count; the gold set, or more context, counts right.*

**DESIGN** — Prioritise **collective evidence coverage** while controlling distractors: structured navigation, summary-based reranking, or iterative retrieval.

**DEPLOYMENT** — ColQwen3 lifts recall at 5 from **58.1% to 81.0%** over BM25, but needs a GPU and 110 ms per query against 0.3 ms. Judge retrieval depth by **end-to-end answers and latency**, not recall alone.

# *Reasoning: Design and Deployment*

*Is the evidence used correctly?*

## Evidence integration

*a diagram, an ablation table*

**Q** If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting?

**Gold** 0.5%

| | | |
|---|---|---|
| **2B** | ✗ | "2.4%" (reads another table) |
| **4B** | ✗ | "−9.6%" (right table, wrong row) |
| **8B** | ✗ | "(75.9 − 74.9) / 75.9 … ≈ 1.32%" |
| ▼ a larger reasoner | | |
| **32B** | ✓ | "75.9 − 75.4 = 0.5" |

*Each size breaks a different step of one chain: locate the table, read the cell, compute.*

## Response calibration

*two tables, two pages*

**Q** How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

**Gold** 16.7%

| | | |
|---|---|---|
| **abstain** | ✗ | "Not answerable" (false refusal) |
| ▼ reason before deciding | | |
| **abstain_cot** | ✓ | "79.9% − 63.2% = 16.7%" |
| **none** | ✓ | "… 16.7 percentage points lower" |
| **cot** | ✓ | "Step 1: … Answer: 16.7%" |

*The question is answerable; the abstain instruction turns multi-step difficulty into refusal.*

**DESIGN** — **Match capacity and inference strategy to the operation.** Separating answer construction from verification helps, at the cost of extra computation.

**DEPLOYMENT** — Quantisation keeps accuracy at a **smaller footprint but is not faster**: under a memory limit prefer a quantised larger model, under a latency limit fewer reasoning steps or a faster backbone.

# *Deployment: Accuracy and Cost*

| Representation | Acc. | Text | Vis. | Tokens | Prefill | Decode |
|---|---|---|---|---|---|---|
| **T** | 29.0 | 499 | 0 | 499 | 1.4 | 5.7 |
| **TL** | 35.7 | 874 | 0 | 874 | 1.8 | 6.6 |
| **TV** | 50.1 | 500 | 459 | 959 | – | – |
| **TLV** | 49.5 | 874 | 459 | 1333 | 26.3 | 6.7 |
| **V** | 42.6 | 20 | 459 | 479 | 29.0 | 7.1 |
| **Retriever** | **R@1** | **R@5** | **R@10** | **Idx. (s)** | **Qry. (ms)** | **GB** |
| BM25 | 32.7 | 58.1 | 68.1 | 0.02 | 0.3 | CPU |
| BGE-M3 | 32.8 | 61.3 | 72.4 | 0.01 | 47 | 1.2 |
| Qwen3-Emb. | 33.5 | 63.9 | 74.8 | 0.19 | 89 | 13.1 |
| ColModernVBERT | 47.1 | 74.0 | 82.7 | 0.91 | 41 | 5.1 |
| ColQwen2.5 | 49.2 | 77.2 | 84.8 | 1.50 | 293 | 8.6 |
| ColQwen3 | 53.7 | 81.0 | 87.9 | 0.57 | 110 | 9.4 |
| **Reasoner** | **T** | **TL** | **TLV** | **V** | **GB** | **Time (s)** |
| Qwen3-VL-8B, bf16 | 31.9 | 38.4 | 52.4 | 45.5 | 17.5 | 30.7 |
| 8-bit | 32.5 | 37.9 | 53.6 | 45.7 | 10.3~ | 43.5 |
| 4-bit | 31.8 | 39.4 | 51.8 | 45.5 | 6.8~ | 35.4 |
| Qwen3-VL-32B, 4-bit | 37.0 | 42.1 | 59.1 | 52.3 | 20.0~ | 47.5 |
| InternVL3-8B | 19.3 | 25.3 | 32.6 | 24.1 | 15.9 | 5.3 |
| MiniCPM-V-4.5 | 30.7 | 38.1 | 60.1 | 53.3 | 17.4 | 47.2 |
| GLM-4.6V-Flash | 25.9 | 32.9 | 42.7 | 30.9 | 20.6 | 19.8 |
| Llama-3.2-11B-V | 26.5 | 34.3 | 33.3 | 25.3 | 21.3 | 10.4 |
| Gemma-3-12B | 30.5 | 36.9 | 48.1 | 41.6 | 24.4 | 8.0 |

PRACTICAL DISCUSSION

## Every choice has a cost side

- **Representation:** embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost about 26 s of prefill against 1.8 s.
- **Retriever:** ColQwen3 reaches 81.0% recall at 5 against 58.1% for BM25, but needs a GPU and 110 ms per query against 0.3 ms.
- **Reasoner:** 4-bit keeps accuracy at a smaller footprint (51.8% in about 6.8 GB) but is not faster.

# *From Findings to Design*

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

### Keep text and images together

Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

### Gather broadly, present selectively

Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

### Check whether the answer is supported

Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

# *Simple Method Cost*

| | | Tokens | | | Latency (s) | | Peak |
|---|---|---|---|---|---|---|---|
| | Turns | Input text | Input visual | Output | Prefill | Decode | VRAM (GiB) |
| `reader` | 11.5 | 5,030 | 816 | 697 | 1.8 | 46.8 | 7.2 |
| `synthesizer` | 1.2 | 9,307 | 2,018 | 463 | 3.8 | 34.7 | 8.1 |
| `arbiter` | 1.2 | 6,806 | 1,734 | 464 | 2.9 | 33.3 | 7.9 |
| Per question, mean | 13.8 | 76,669 | 13,767 | 9,092 | 28.5 | 617.5 | 8.2 |
| Per question, median | 9.0 | 43,113 | 9,830 | 6,078 | 15.8 | 385.0 | 8.0 |

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.